Long-Context Compression & Linear-Attention Memory

Research progress — Papers A / B / C

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RCA Foundation Model / Long-Context Memory

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All artifacts hyperlinked to github.com/BDeMo/research-notes (blue = clickable).

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1. Background: the long-context memory wall

Problem. Long inputs carry documents, tools, history, and unfinished agent work, but processing every token requires more compute and memory as context grows.

- A larger window does not guarantee that the model uses the needed evidence; position, distractors, and reasoning depth still change the answer.
- **Two practical responses:**
  1. *Select raw evidence* — retrieval, prompt compression, or a bounded window.
  2. *Build a compact state* — learned soft memory or a recurrent linear-attention state.
- **System requirement:** preserve a raw path when the compact state is insufficient.

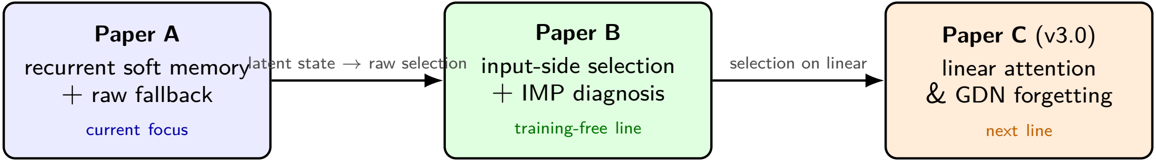
Guiding question

On a *frozen* base, what should be kept as compact memory, and when should an agent return to raw evidence?

2. One thesis, three papers (the mainline)

Thesis

Long-context memory is an **interface and routing** problem: a compact state is useful only where the next decision survives compression, and raw evidence must remain available outside that region.



Paper A asks when a learned short state is usable; Paper B asks which raw evidence should be retained without training; Paper C studies how the same selection problem changes for recurrent linear-attention memory.

3. Paper B — the diagnosis (model-invariant)

Systematic full-test map across 9 model families (1.7B–14B, Qwen2.5/3/3.5, GLM-4, Ministral, Llama; quad + linear):

- **No universal compressor** — the winner flips by task (retrieval / extractive / MC / reasoning).
- **"More context can hurt"** — on literary-MC *full* $<$ blind guessing (QuALITY full 7.2 $<$ 17.9); compression *denoises*.
- **Retrieval $\gg$ full** on ultra-long (∞Bench) — selecting evidence beats reading everything.
- **Model-invariant**: the crossover & denoising hold on *every* family, incl. cache-free linear GDN.

This diagnosis — not a single number — is Paper B's durable contribution. fact base (F27/F31/F43) full-test main table

4. Paper B — the method: IMP (Importance-routing Prefilter)

Training-free, input-side, architecture-agnostic prefilter on a *frozen* base:

1. Score context tokens/chunks by cheap $O(L)$ signals (lexical BM25 + query-relevance + surprisal).
2. Keep the top spans/chunks *verbatim* to a budget; drop the rest. Feed to the frozen base.
3. **Route** the granularity per input (`auto`): long/extractive → chunk-BM25 over the *full doc*; literary-MC-that-fits → token-importance spans.

- No weights trained; runs on dense *and* linear bases; no KV cache needed.
- Covers regimes that RAG (fails reasoning) and KV-eviction (can't run on linear) each only half-cover.

method spec + code path   implementation

5. Paper B — full results (Qwen3-8B, all methods × benches)

bench	full·t	rag	kvzip‡	knorm‡	tome	**auto (ours)**	best
∞Bench (>100k)	52.8	64.2	35.8	25.8	51.5	**70.3**	**ours**
RULER-NIAH	99	94	**99**	85	0	98	kvzip/ours*
squad_v2	65.9	67.2	64.9	16.5	38.9	66.9	rag≈ours
trivia_qa	70.9	72.7	71.2	40.7	62.3	**72.8**	**ours**
lb_multifieldqa	38.6	38.8	37.1	30.7	–	**39.5**	**ours**
lb_narrativeqa	14.0	13.7	11.2	9.3	13.5	**14.3**	**ours**
hotpot_qa	57.2	55.8	56.2	23.1	43.9	48.8	full
QuALITY-hard	9.2	10.5	**29.9**	29.6	11.7	12.9	**kvzip**‡
MuSR	58.9	56.7	55.6	53.3	48.9	48.9	full

·t = truncated to 16k;

‡ = needs KV cache (*cannot* run on linear). ours = `auto`. complete table (16 benches, 9 families)

6. Paper B — honest standing vs *all* baselines (not just RAG)

Where ours wins / ties

- **Wins**: ∞Bench, trivia, multifieldqa, narrativeqa.
- **Ties best**: squad, lb_hotpotqa, ms_marco, LongBench-v2; best *arch-agnostic* on RULER.
- **Only method that runs on linear** among the strong ones.

Where we honestly lose

- Literary-MC (QuALITY) $\rightarrow$ *KV-eviction* (kvzip/knorm $\approx$30) — but they need a KV cache.
- Dense reasoning (MuSR, musique) $\rightarrow$ *full* (needs everything).

Framing for the paper

Contribution = the **diagnosis** + a **regime-spanning training-free router**, *not* "beats everything". We state plainly where KV-eviction / full win. best-of-all comparison

7. Paper B — rigor: a bug we found & fixed

- **Symptom:** IMP looked -19 vs RAG on ∞Bench.
- **Root cause (case-level audit):** IMP *truncated* the context to 16k *before* selecting; RAG retrieved over the *whole* 131k-token doc. **87% of RAG's kept content came from beyond 16k — IMP was structurally blind to it.**
- **Fix:** chunk-family retrieves over the full doc (no forward needed). ∞Bench 51 $\rightarrow$ **76** — the gap *reversed* to a win.
- **Discipline installed:** every result labels `full`·`trunc16k`; per-bench truncated-vs-true-full audited; only ∞Bench materially affected.

correctness audit case-level diff code review

8. Paper B — Occam simplification $\to$ v3.0

What actually drives the wins? An ablation of the (baroque) `auto` router shows:

- The **chunk branch = full-doc BM25 retrieval** carries nearly all the wins.
- Importance signals (surprisal, query-dot) add ≈ 0; the span branch only matters in literary-MC (where it still loses to KV-eviction).

v3.0 = `IMP-lite`: one mode, one signal, one knob

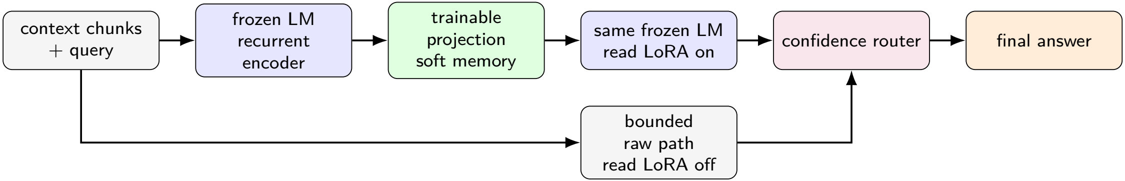
Full-document lexical chunk retrieval on a frozen base — no span mode, no surprisal, no router, no training. Occam: drop every non-load-bearing part; keep the workhorse. v3.0 design + validation plan

Open question (decide empirically): is IMP-lite meaningfully > RAG, or is it "RAG configured for compression + the diagnosis"?

9. Paper A — memory is a routed interface

Question

When can an agent answer from a short learned state, and when should it return to raw evidence?



- One shared backbone supplies encoding, memory reading, confidence, and raw fallback.
- The contribution is the **complete compact-versus-raw route**, not compression ratio alone.
- The formal fixed-threshold test currently certifies no nonzero coverage; routing results are held-out empirical analysis.

paper story   all-benchmark tables

10. Paper A — core results (three seeds)

base	method	QuALITY	BFCL	HotpotQA
Qwen3-8B	Raw	7.2	92.4	53.7
	SFT	81.7±1.7	95.4±0.3	68.8±0.6
	Window	15.7	55.7	26.2
	LLMLingua-2	14.3	70.3	22.1
	Compressor (w/o gate)	54.4±0.2	72.3±0.5	28.9±0.2
	Compressor (w/ gate)	**54.6**	**88.5**	**50.9**
Qwen3.5-9B	Raw	7.1	84.5	53.9
	SFT	85.0±0.4	94.9±1.0	71.7±0.6
	Window	16.7	52.8	24.8
	LLMLingua-2	20.3	60.8	28.9
	Compressor (w/o gate)	**51.5±1.7**	72.0±0.8	30.5±0.3
	Compressor (w/ gate)	51.4	**80.5**	**51.7**

- The compressor without the gate is the strongest completed compact path across the displayed core tasks; SFT is a full-cost reference.
- The compressor with the gate recovers much of the raw gap on BFCL and HotpotQA; QuALITY already favors compact memory.
- SQuAD-v2 is retained only as a short-passage exact-text diagnostic, not a headline benchmark.

audited core table routing metrics

11. Paper A — real long-context transfer (seed 42)

base	method	LongBench-v2	∞Bench-choice	BABILong
Qwen3-8B	No context	33.4	41.9	2.0
	Raw	31.2	52.8	18.1
	Compressor (w/o gate)	20.1	21.4	0.9
	Compressor (w/ gate)	**31.4**	**52.8**	**18.1**
Qwen3.5-9B	No context	27.0	42.4	3.1
	Raw	33.0	52.4	10.5
	Compressor (w/o gate)	22.5	31.9	2.2
	Compressor (w/ gate)	**33.4**	**52.4**	**11.9**

New result: the boundary is task-structured

On multi-hop LongBench transfer, the compressor without the gate can beat truncated raw (Qwen3 HotpotQA: 19.1 → 30.5; Qwen3.5: 3.9 → 26.6), and the gate improves it further. On single-document / extractive targets, raw wins and the gated variant falls back. Across the harvested two-base targets, **Compressor (w/ gate)** $\geq$ **Raw in every cell**.

Single seed; error bars pending. RULER and ∞Bench K32 repairs remain.

full long-context breakdown

12. Paper A — full execution status (Wed 2026-07-22 06:33 PT)

stage	done	run	fail	pending
core main	88	0	0	0
replicate + SFT audit	9	0	0	0
transfer adapters	46	0	2	0
real long-context	50	3	2	63
8-model generality	11	1	0	36
budget / RULER	0	0	0	23
ablation	0	0	0	36
official baselines	0	0	0	52

Live workers

Three Qwen3.5-4B long-context jobs + one Qwen3.5-9B generality job. One GPU on the generality node is unavailable.

Completed evidence

- Core grid, output audit, gate analysis, replicate, SFT reaudit.
- Two-base long-context harvest: 44 cells.

Repairs / blockers

- K32 ∞Bench adapters failed on both main bases.
- GLM-4 and Qwen3.5-9B K32 source-adapter repairs remain.
- LCLM / Semi-Dynamic cloned; official execution not started.

Execution order

1. continue long-context across the remaining bases;
2. complete generality;
3. budget/RULER $\rightarrow$ ablation/cost;
4. official baselines.

benchmark + baseline inventory experiment receipt

13. Paper C (v3.0) — why linear attention; the research line

Focus (next line): linear attention & its variants (Qwen3.5/3.6 GDN family; *no* Qwen3 control).

- **The tension:** a *fixed-size state* is efficient but has a structural **recall / forgetting bottleneck** (recall–throughput tradeoff; state-tracking limits).
- **Our hook (already shown):** on Qwen3.5-GDN the long-context diagnosis reproduces, but **KV-eviction can't even run** (no KV) — only input-side selection (IMP-lite/RAG) works.
- **The opening:** on linear bases the problem isn't "shrink the KV" (there is none) — it's **feed the right small context / augment the state's recall**.

Candidate spine

C-1 diagnose GDN forgetting (no training) + C-2 training-free input-side memory (IMP-lite) that KV-methods structurally can't provide. research line

14. Paper C — the landscape (delta-rule family)

model	decay	erase	write	delta
Mamba-2	scalar	—	scalar	no
Gated DeltaNet (our base)	scalar	tied scalar	tied scalar	yes
KDA / Kimi-Linear	channel-wise	tied scalar	tied scalar	yes
GDN-2 (NVIDIA, 2026)	channel-wise	**channel-wise**	**channel-wise**	yes

- Frontier = **how to *edit* the compressed memory** (erase vs write) without scrambling associations.
- Field findings: the *delta rule* dominates recall; **GDN-2's erase gate carries most of its gain** (selective forgetting of stale keys is the lever); channel-wise helps long-range but can hurt precise recall.

reference collection (11 categories, arXiv)

Training-free anti-forgetting modules tested on Qwen3.5-GDN (real benches):

bench	full	imp-lite	rel-last	replay	rag
∞Bench	55	76	75	**78**	65
lb_multifieldqa	38.6	42.1	40.7	**44.0**	41.1
babilong-qa1	65	**68**	62	61	54

- **Input-side compression rescues GDN on ultra-long** (∞Bench 55→76) — don't overflow the fixed state.
- Recency-reorder modules (rel-last / replay) = *marginal, task-dependent* ⇒ Occam: imp-lite default.
- Honesty: our synthetic single-needle probe was *inconclusive* (task too easy / a probe bug we caught); the trustworthy evidence is the *real-bench* grid above. 128k probe OOMs (GDN kernel).

ablation design module results (+ probe caveats)

16. Development logic & timeline

1. **Paper A** — recurrent soft memory + raw fallback: core grid complete; two-base long-context transfer harvested; remaining scale, budget, ablation, and official baselines running.
2. **Paper B** — complementary input-side diagnosis + IMP selection on a frozen base; full-test audit and Occam simplification completed.
3. **Shared result** — no representation wins everywhere; task structure determines whether compact memory, selected raw spans, or bounded raw context should answer.
4. **Paper C** — carry the selection question to recurrent linear-attention memory, where state forgetting replaces raw-window growth as the main bottleneck.
5. **Current priority** — finish Paper A's remaining grids and lock its complete result tables before expanding Paper C.

17. Cross-cutting insights & next steps

Insights

- Compact memory has a **task-structured operating region**: strong on some decision and multi-hop interfaces, weak on exact planted evidence.
- **No universal context path**; the agent should choose among learned memory, selected raw evidence, and bounded raw context.
- Reliability must be evaluated jointly with coverage and total cost, not only average compressed accuracy.

Next steps

- **Paper A:** complete transfer / generality; run RULER, budget, ablation, cost, and official-native baselines; then lock tables and figures.
- **Paper B:** run the IMP-lite Occam ablation; finalize v3.0; decide the RAG-delta claim.
- **Paper C:** secure the in-band **Qwen3.6-27B** (hybrid) checkpoint; RULER access (HF); diagnose GDN forgetting on real long-context; test input-side vs erase-gate-retrofit fixes.

18. Artifact index (all links to GitHub)

Paper B consolidated report · complete results · main table · fact base · best-of-all correctness audit · code review · v3.0 method · 5-day summary

Paper A paper draft · all tables · benchmark inventory · baseline plan · experiment receipt

Paper C overview · research line · references · GDN forgetting design · module results

Repo: github.com/BDeMo/research-notes